Calculus

Differentiation:

- $\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$
- $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\frac{du}{dx} + u\frac{dv}{dx}}{v^2}$
- $\frac{d}{dx}(x^n) = n \cdot x^{n-1}$
- $\frac{d}{dx}(a^x) = a^x \log a$
- $\frac{d}{dx}(\log x) = 1/x$
- $\frac{d}{dx}(\sin x) = \cos x$, $\frac{d}{dx}(\cos x) = -\sin x$
- $\frac{d}{dx}(\tan x) = \sec^2 x$
- $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$
- $\frac{d}{dx}(\sec x) = \sec x \cdot \tan x$
- $\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cdot \cot x$
- $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$
- $\frac{d}{dx}(\cot^{-1} x) = \frac{-1}{1+x^2}$
- $\frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$ → $\operatorname{cosec}^{-1} x = \frac{-1}{x\sqrt{x^2-1}}$

Integration

- $\int x^n = x^{n+1}/n+1$
- $\int \cos x\,dx = \sin x + C$
- $\int \sin x\,dx = -\cos x + C$
- $\int \sec^2 x\,dx = \tan x + C$
- $\int \operatorname{cosec}^2 x\,dx = -\cot^2 x$
- $\int \sec x \tan x\,dx = \sec x$
- $\int \operatorname{cosec} x \cot x\,dx = -\operatorname{cosec} x + C$.
- $\int \frac{1}{\sqrt{1-x^2}}dx = \sin^{-1} x = -\cos^{-1} x$
- $\int \frac{1}{\sqrt{1+x^2}}$ → $\int \frac{1}{1+x^2}dx = \tan^{-1} x = \cot^{-1} x$
- $\int a^x = a^x/\log a$
- $\int \frac{1}{x\sqrt{x^2-1}} = \sec^{-1} x = \operatorname{cosec}^{-1} x$
- $\int 1/x = \log x$
- $\int \tan x = \log|\sec x| + C$
- $\int \cot x = \log|\sin x| + C$ or $-\log|\operatorname{cosec} x|$
- $\int \sec x = \log|\sec x + \tan x|$
- $\int \operatorname{cosec} x = \log|\operatorname{cosec} x - \cot x|$.

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Mean Value theorem:

✓ Lagrange's theorem:

→ $f(x)$ is Conti^{n} $[a,b]$

→ diff in (a,b)

$$\text{then } f'(c) = \frac{f(b)-f(a)}{b-a}$$

✓ Rolel's theorem:

→ $f(a) = f(b)$

$$f'(c) = 0$$

✓ Limit:

$\frac{0}{0}, \frac{\infty}{\infty}$ → only valid Condition for L'Hospital.

→ $$\lim_{x \to a} \frac{x^n - a^n}{x-a} = n \cdot a^{n-1}$$

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→ Consistent state: Have one or more sol^{n}
→ Non-Consistent: No solution.

✓ Linear Equations

- **Homogeneous ($AX = 0$)**
  1. → $X = 0$ is always sol^{n}. [Trivial sol^{n}]
  2. → $\rho(A)$ = # Variables (rank) [Unique sol^{n}]
  3. → $\rho(A) <$ # Variable [Infinite sol^{n}]

- **Non-Homogeneous ($AX = B$)**
  1. → $\rho[A:B] \neq \rho[A]$ → No sol^{n}
  → $\rho[A:B] = \rho(A) \neq$ # Unknowns → Infinite sol^{n}.
  → $\rho[A:B] = \rho(A) =$ # Unknowns → Unique sol^{n}.

→ If the rank of the matrix of the given vector is less than the no of vectors, then the vectors are linearly dependent.

✓ LU Decomposition - $A = LU$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ L_{21} & 1 & 0 \\ L_{31} & L_{32} & 1 \end{bmatrix} \quad U = \begin{bmatrix} U_{11} & U_{12} & U_{13} \\ 0 & U_{22} & U_{23} \\ 0 & 0 & U_{33} \end{bmatrix}$$

or

$$\begin{bmatrix} L_{11} & 0 & 0 \\ L_{21} & L_{22} & 0 \\ L_{31} & L_{32} & L_{33} \end{bmatrix} \begin{bmatrix} 1 & U_{12} & U_{13} \\ 0 & 1 & U_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

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Linear Algebra

$|AA^T = I|$ → orthogonal Matrix.

✓ $A(\text{adj}A) = |A| \cdot I$

$|A(\text{Adj}A)| = ||A||. \Rightarrow \boxed{|\text{Adj}A| = |A|^{n-1}}$

✓ $\boxed{|\text{adj}(\text{adj}A) = |A|^{n-2} \cdot A|}$ ✓ $\text{adj}(AB) = \text{adj}(A) \cdot \text{adj}(B)$

✓ $\boxed{|\text{Adj}(kA) = k^{n-1}\,\text{Adj}(A)|}$

✓ Eigen Values: $|A - \lambda I| = 0$.

→ EV of Real symmetric & Hermitian Metric is real.

→ EV of Real skew symmetric & skew Hermetian metric is either pure imaginary or zero.

→ EV of orthogonal & unitary matrix is $|\lambda| = 1$ [Unit Modulus]

→ λ_1, λ_2 ___ then for → $kA = k\lambda_1, k\lambda_2, --$
→ $A^{-1} = 1/\lambda_1, 1/\lambda_2, --$
→ $A^k = \lambda_1^k, \lambda_2^k, --$

→ EV of A = EV of A^T.

→ Sum of eigen value = Trace [sum of dignal ele]

→ Product of EV = $|A|$.

→ If A & B are two matrix of same order then
EV of AB = EV of BA.

→ The Eigen Vector Corres. to different value of a real symmetric matrix are orthogonal to each other & dot product would be zero.

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✓ Expectation, Variance, Co-variance.

→ Covariance:

$$\boxed{Cov(X,Y) = E(XY) - E(X)\cdot E(Y)}$$

→ Expectation:

- → $E[aX+b] = aE[X]+b.$
- → $E(X+Y) = E(X)+E(Y)$
- → $E(aX) = aE(X)$
- → $E(XY) = E(X)\cdot E(Y)$

→ Variance:

- → $Var(X) = E(X^2) - [E(X)]^2$
- → $Var(aX) = a^2 Var(X)$
- → $Var(b) = 0.$ ⇒ Variance of Constant is zero.

Independent

→ $Var(X+Y) = Var(X) + Var(Y)$

not Independent

→ $Var(X+Y) = Var(X) + Var(Y) + 2Cov(X,Y)$

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Probability

Standard Deviation : $E(X^2) - [E(X)]^2$

→ Binomial Random : $(n, p) \Rightarrow {}^nC_r \, p^r (1-p)^{n-r}$ (p → Prob of success)

- ↳ Mean : np
- ↳ Variance : npq

→ Poission : $\frac{e^{-\lambda} \cdot \lambda^r}{\lfloor r}$ $\boxed{\lambda = np}$ → Mean.

- ↳ Variance : λ.

$\boxed{S.D = \sqrt{\text{Variance}}}$

→ Uniform : $\begin{cases} \frac{1}{b-a} & a < x \le b. \\ 0 & ; \text{otherwise} \end{cases}$

- ↳ Mean : $\frac{a+b}{2}$
- ↳ Variance : $\frac{(b-a)^2}{12}$

→ Exponential : $\begin{cases} \lambda e^{-\lambda x} & ; x \ge 0 \\ 0 & ; \text{otherwise} \end{cases}$

Mean : $1/\lambda$

Variance : $1/\lambda^2$

→ Normal Distribution : $\frac{1}{\sqrt{2\pi}\,\sigma} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$

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